

MSc Software Engineering graduate from Heriot-Watt University and product engineer with experience across open source maintenance, product engineering, data analysis, internal tools, and game systems. Strong at turning ambiguous requirements into working systems, communicating across functions, and building useful prototypes with a pragmatic engineering approach.

Key skills: TypeScript, React, React Three Fiber, Node.js, JavaScript, Python, Java, C++, PHP, MVC architecture, boardgame.io, data analysis, prototyping, business reviews, cross-functional delivery.

EXPERIENCE

Open Source Maintainer - boardgame.io Remote TypeScript game-state framework - Maintain an open source TypeScript framework used to model deterministic, turn-based game state. - Review issues, pull requests, documentation gaps, and framework behavior with attention to API stability. - Apply maintainer experience directly in Empires of the Skies, a multiplayer strategy game built on boardgame.io.	May 2024 - Present
Engineer & Catalyst - CEO's Office Bizom (Mobisy Technologies Ltd), Bengaluru, India - Integrated BNPL with Bizom's SFA, improving workflow efficiency and user experience. - Hosted weekly company-wide business reviews and town halls to improve cross-functional visibility. - Owned and promoted the suggested-order feature, improving ordering accuracy and customer outcomes.	Apr 2023 - Jul 2024
Data Scientist - CEO's Office Bizom (Mobisy Technologies Ltd), Bengaluru, India - Developed churn analysis tooling to support customer retention strategy and product decisions. - Built CEO Office experiments including Catalog as a Service and DAU-focused prototypes. - Improved operational efficiency through practical prototypes and analysis-backed recommendations.	Jun 2022 - Dec 2022
Software Engineer - Platform Bizom (Mobisy Technologies Ltd), Bengaluru, India - Owned alert infrastructure and resolved high-priority platform tickets. - Maintained and rewrote legacy code to improve runtime and reliability. - Reused alert technology to support cron operations and internal system workflows.	Aug 2021 - Jun 2022
Software Engineer Intern - Analytics Bizom (Mobisy Technologies Ltd), Bengaluru, India - Generated client reports, dashboards, and alerts for data-driven decision-making. - Built and maintained migration tooling for legacy platform reports. - Contributed to analytics reliability through report rewrites and internal tooling support.	Aug 2021 - May 2022

EDUCATION

MSc Software Engineering Heriot-Watt University, Edinburgh, UK	Jun 2026	B.E. Information Science & Engineering Visvesvaraya Technological University, Bengaluru, India	Jan 2022
Diploma in Game Development Backstage Pass Institute of Gaming	Jan 2021	High School, PCM + CS Delhi Public School (DPS)	Jan 2016

PROJECTS

- Empires of the Skies - Digital strategy board game and AI opponent | TypeScript, React, Node.js, Python, boardgame.io**
- Architected a full-stack multiplayer web application by migrating a legacy codebase into a three-package monorepo.
 - Implemented a 6-player strategy engine with server-authoritative state, 69 move types, and a 7-phase round structure.
 - Built a hybrid AI using 26 heuristic evaluators with Monte Carlo Tree Search, averaging 25.6 ms decision time.
 - Created a CMA-ES parameter tuning pipeline in Python, optimizing 177 parameters for a 15x AI quality improvement.
- Nimina's Rhythm - Team rhythm game | Unity, C#**
- Designed a conductor system to sync jumps, dashes, attacks, platforms, hazards, and enemies to track BPM.
 - Implemented precise input-validation windows for beat-based interactions.
- Coffee Shop Simulation | Java, JavaFX, Multithreading, Design Patterns**
- Built an MVC simulation with model classes, JavaFX UI, Observer updates, and Singleton file logging.
- Student Management System | C++, Inverted List Indexing**
- Engineered a modular multi-file codebase and implemented an inverted index for fast student record searches.

ADDITIONAL UK EXPERIENCE

Front of House / Hospitality Edinburgh, UK	Sept 2024 - Present
- Managed high-volume customer interactions in a fast-paced environment while completing a full-time Master's degree. - Demonstrated cross-cultural communication, conflict resolution, adaptability, and UK workplace experience.	

CERTIFICATES

- | | |
|--|--|
| <ul style="list-style-type: none">Microservice Architectures (Coursera), Jan 2024 - VSJWQFSXG3KWBuilding HTTP APIs with Spring (Coursera), Dec 2023 - GSR9V996RVN4Advanced SQL (HackerRank), Jan 2024 - 46ef42734ce9 | <ul style="list-style-type: none">Intermediate SQL (HackerRank), Dec 2023 - 1a8054aec384C Programming, Intermediate (HackerRank), Sep 2020 - f0a51d468d78Learning How to Learn (Coursera), Nov 2020 - RDNWL4NF9WCZ |
|--|--|